ST MARY'S DSG, KLOOF

GRADE: 10 **5 August 2025**

PHYSICAL SCIENCE CLASS TEST

TIME: 42 MINS **TOTAL: 43**

ASSESSOR: G. Main **MODERATORS: K. Bolton & T. Kirsten**

NAME: Memo

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 6 pages. A data sheet is provided.
2. Answer on the question paper in the spaces.
3. Write your name on your question paper.

QUESTION 1 – MULTIPLE CHOICE

For the next three questions you must make a choice out of the answers A - D given.
On the grid provided, make a cross over the letter you think is most correct, e.g.

1.4	A	~~B~~	C.	D

1.1	A	B	C	~~D~~
1.2	~~A~~	B	C	D
1.3	A	~~B~~	C	D

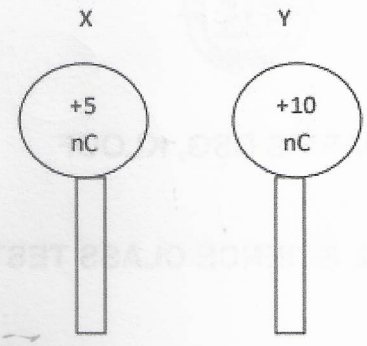
1.1 Which statement explains why a PVC rod becomes negatively charged when rubbed with a woollen cloth.

A Protons are transferred from the PVC rod to the woollen cloth.
B Protons are transferred from the woollen cloth to the PVC rod.
C Electrons are transferred from the PVC rod to the woollen cloth.
D Electrons are transferred from the woollen cloth to the PVC rod.

1.2 The strips on a charged gold leaf electroscope deflect more when a charge is brought close. The approaching charge is:

A a like charge to that of the charged electroscope.
B an unlike charge to that of the charged electroscope.
C a positive charge.
D a negative charge.

1.3 The diagram below shows two charged pith balls X and Y. X has a charge of +5 nC and Y has a charge of +10 nC. The pith balls are allowed to touch and are then separated. Which of the following statements is true?



A Ball X lost electrons and has a charge of +10 nC after separation.
B Ball X lost electrons and has a charge of +7,5 nC after separation.
C Ball Y lost electrons and has a charge of +5 nC after separation.
D Ball Y lost electrons and has a charge of +7,5 nC after separation.

[6]

QUESTION 2

A neutral plastic ruler becomes charged when it is rubbed with a woollen cloth. After rubbing, the ruler has a charge of $+3{,}44 \times 10^{-15}$ C.

2.1 Distinguish between a neutral object and a charged object. (2)

Neutral no. of e's = no. of protons ✓
Charged no. of e's ≠ no. of protons ✓

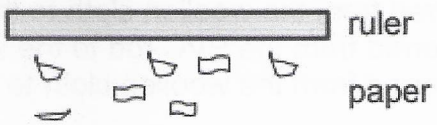
2.2 Does the ruler GAIN or LOSE electrons? (1)

LOSE ✓

2.3 Calculate the number of electrons transferred during the process of rubbing. (3)

$n = \frac{Q}{q_e}$ ✓ $= \frac{3{,}44 \times 10^{-15}}{1{,}6 \times 10^{-19}}$ ✓ $= 21\,500$ ✓

2.4 The charged ruler is now brought closer to pieces of paper. The pieces of paper are attracted to the ruler, as shown below



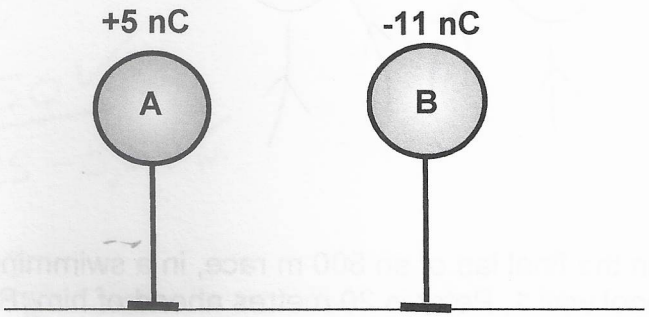
Explain why the pieces of paper are attracted to the ruler. (3)

The electrons in the paper are attracted to the surface of the positive ruler. Polarisation occurs in the paper. The negative side of the paper is attracted to oppositely charged ruler and move towards it.

[9]

QUESTION 3

Two charged spheres **A** and **B** are placed with their centres 4 cm apart.
A carries a charge of **+5 nC** and B carries a charge of **–11 nC**.



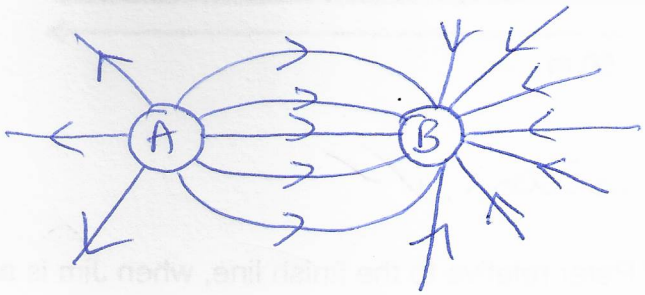
3.1 Define an *electric field*. (2)

A region of space in which an electric charge experiences a force.

3.2 Is there a force attraction or repulsion between the spheres? (1)

attraction

3.3 Draw the electric field pattern between **A** and **B**. (3)



pattern
arrows
more lines around B

3.4 The two bodies are allowed to touch. Will electrons move from A to B or from B to A? (1)

B → A

3.5 Calculate the new charge on objects A and B after they have touched and then separated again. (2)

$$Q_{new} = \frac{Q_1 + Q_2}{2} = \frac{5 + (-11)}{2} = -3\ nC$$

3.6 Calculate the number of electrons that were transferred when the spheres touched. (3)

$\Delta Q = Q_{final} - Q_{int}$

$= (-3) - (+5)$

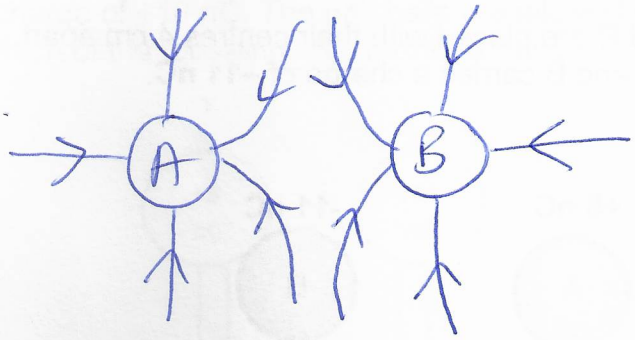
$= -8\ nC$

$n = \frac{\Delta Q}{q_e}$

$= \frac{-8 \times 10^{-9}}{-1,6 \times 10^{-19}}$ conv.

$= 5 \times 10^{10}$

3.7 Draw the new electric field pattern between A and B after they have touched and separated. (3)

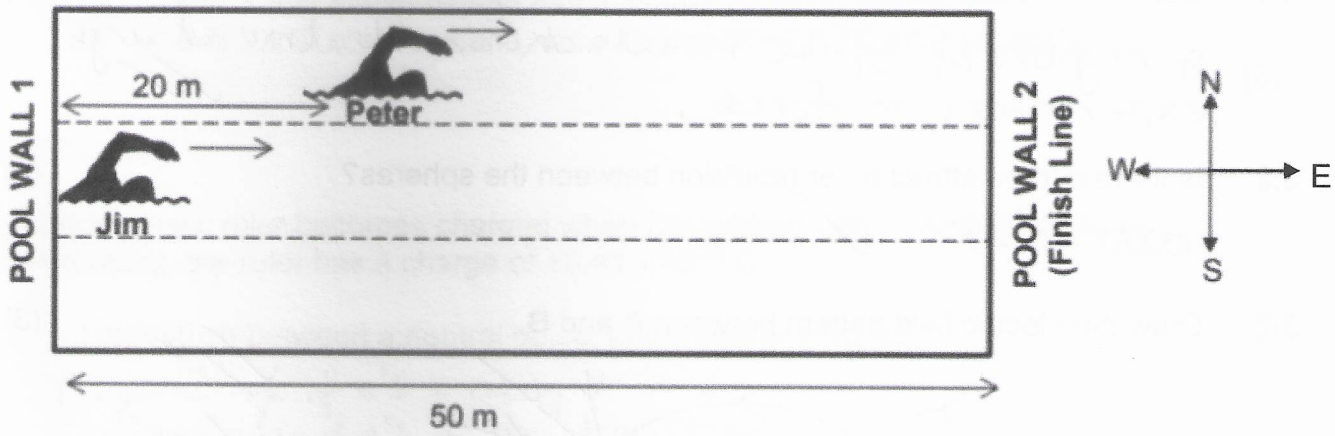


✓ diamond
✓ arrows
✓ same no. of lines

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QUESTION 4

Peter and Jim are both on the final lap of an 800 m race, in a swimming pool of length 50 metres. When Jim is at pool wall 1, Peter is 20 metres ahead of him. Both swimmers are moving eastwards to finish the race at pool wall 2.



4.1 Define *displacement*. (2)

Change in position. ✓✓

4.2 What is the position of Peter relative to the finish line, when Jim is at pool wall 1. (2)

30 ✓m west ✓

4.3 What is the position of Peter relative to Jim, when Jim is at pool wall 1. (2)

20 ✓m east ✓

4.4 Peter swims at an average velocity of 1,2 m.s^{-1} east. Calculate the time taken for Peter to reach the finish line from his current position. (3)

$v = \frac{\Delta x}{t}$

$1,2 = \frac{30}{t}$ ✓ ✓ (col. 4.2)

$t = \frac{30}{1,2}$

$= 25$ s ✓

4.5 Calculate the average speed at which Jim needs to swim in order to finish the race 0,5 s ahead of Peter. (4)

$$v = \frac{\Delta x}{t}$$

$$= \frac{50}{25 - 0,5}$$

$$= \frac{50}{24,5}$$

$$= 2,04 \text{ m.s}^{-1}$$

[13]